

Open tools for more accessible and transparent biologically informed deep learning

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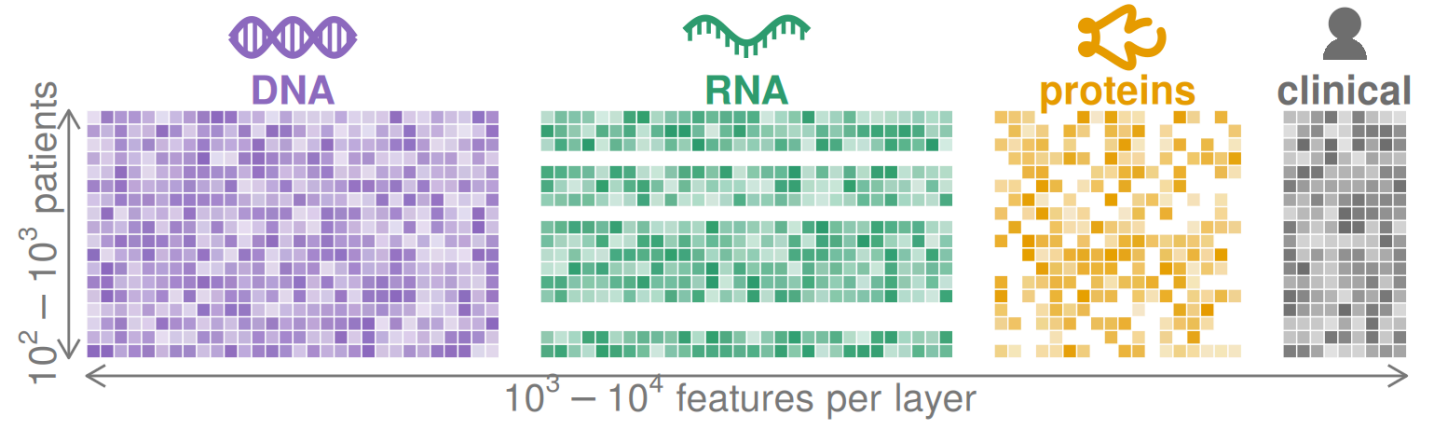
Why prior knowledge?



German Research Center for Artificial
Intelligence · Data Science and its Applications



Cluster for Atherothrombosis and Individualised
Medicine · Mainz



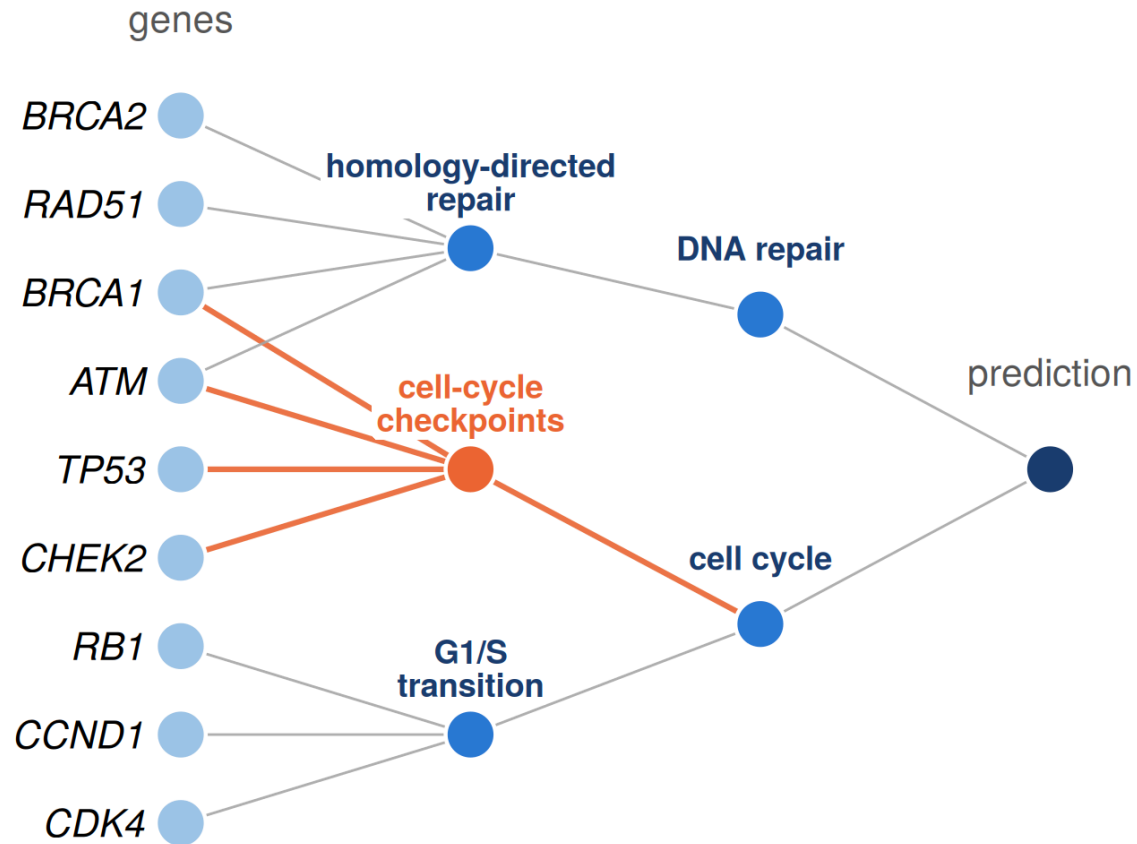
Gefördert durch:



small · noisy · high-dimensional · multimodal

→ borrow strength from **prior knowledge**

Wiring networks like biology



inputs, per patient and gene:
mutated? copy number? activity?

pathway: genes whose products
work together on one cellular job ·
curated (Reactome, GO) · nested

BINN = keep only the connections the
database supports

→ **predict + explain**

P-NET (*Nature* 2021) · DCell (2018) · GenNet (2021) ·
BINN (2023)

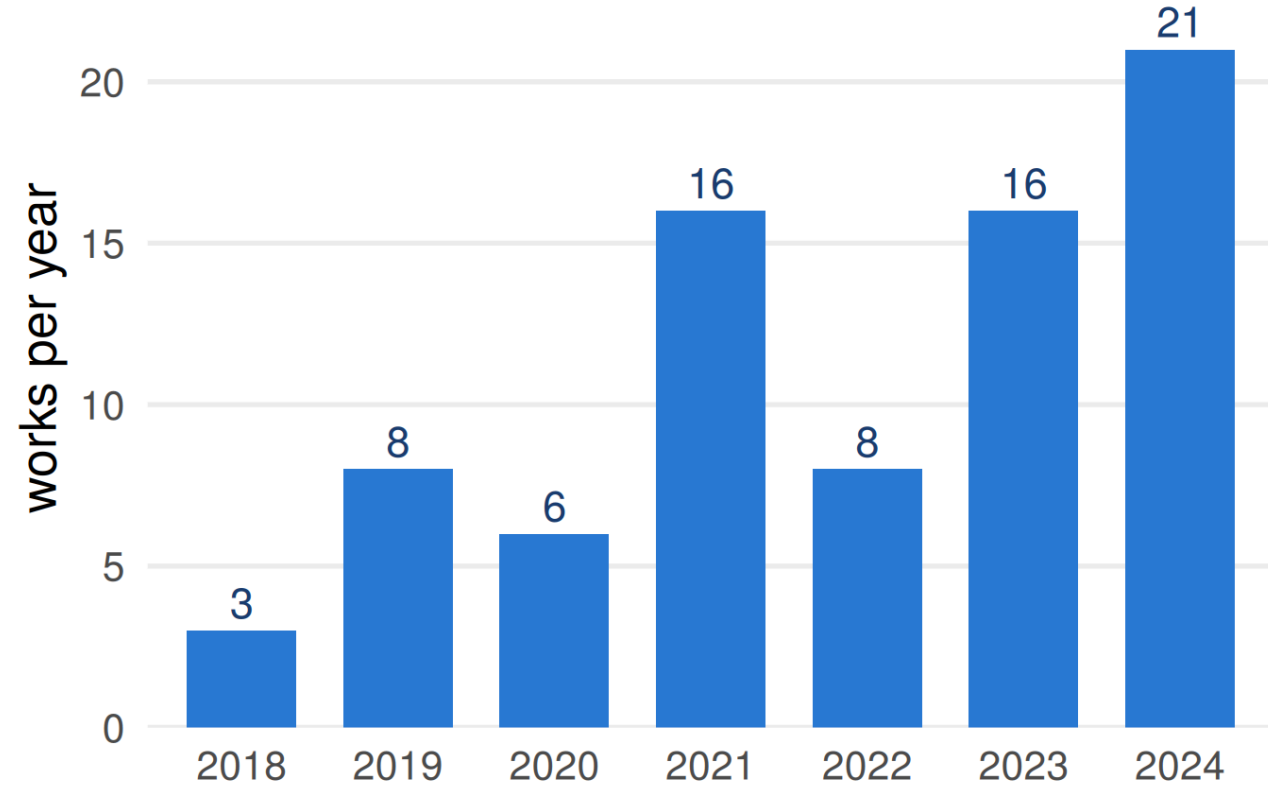
A crowded field

86

papers, 2018–2024

Selby et al., *Front. AI* 2025

- no shared **terminology**
- no shared **tools**
- no shared **benchmarks**



works listed in the review's Table A.1 (n = 78), by year

Under scrutiny

question	study	finding
replicable?	Pedersen et al. 2023 · <i>Cancer-Net</i>	P-NET reproduced; shuffled wiring worse
stable?	Esser-Skala & Fortelny 2023	importance varies across retrainings, tracks network position
does biology matter?	Caranzano et al. 2026 · “ <i>Sparsity is all you need</i> ”	20 models: random $\approx$ real
what can it learn?	Miller et al. 2026 · simulation	limited by n , signal, sparsity
how many answers?	Gankin et al. 2026 · 101 replicates	explanations correlate only ≈ 0.5 between reruns

Same model, opposite answers

P-NET, shuffled wiring

AUPR $\approx 0.89 \rightarrow \approx 0.83$

biology matters

Pedersen et al. 2023

P-NET, random wiring

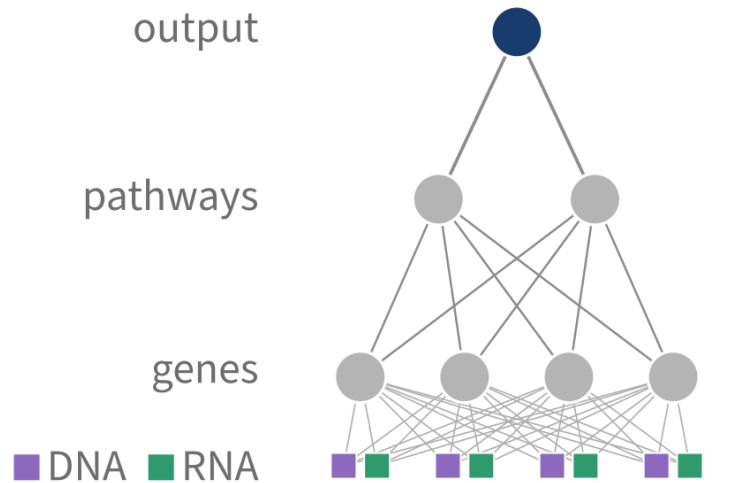
AUC $0.899 \rightarrow 0.896$

sparsity is all you need

Caranzano et al. 2026

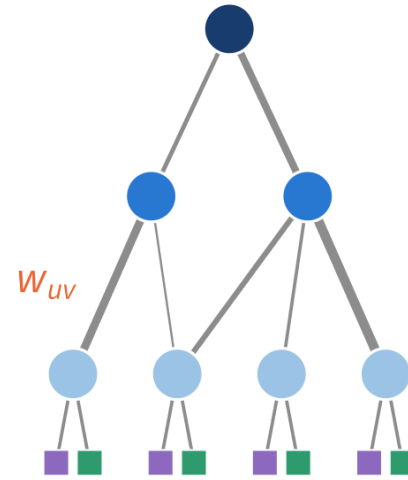
Right question? Right tool?

A BINN is message passing on a graph



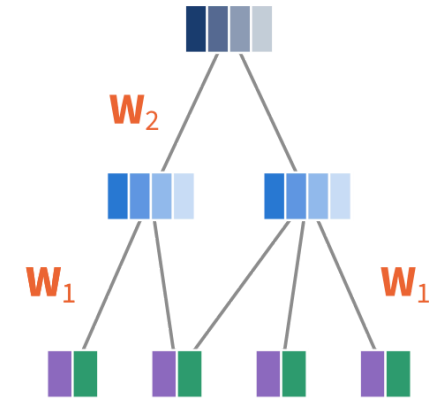
fully connected

no graph;
hidden units anonymous



BINN

graph = mask;
one number per node,
own weight per link



GNN

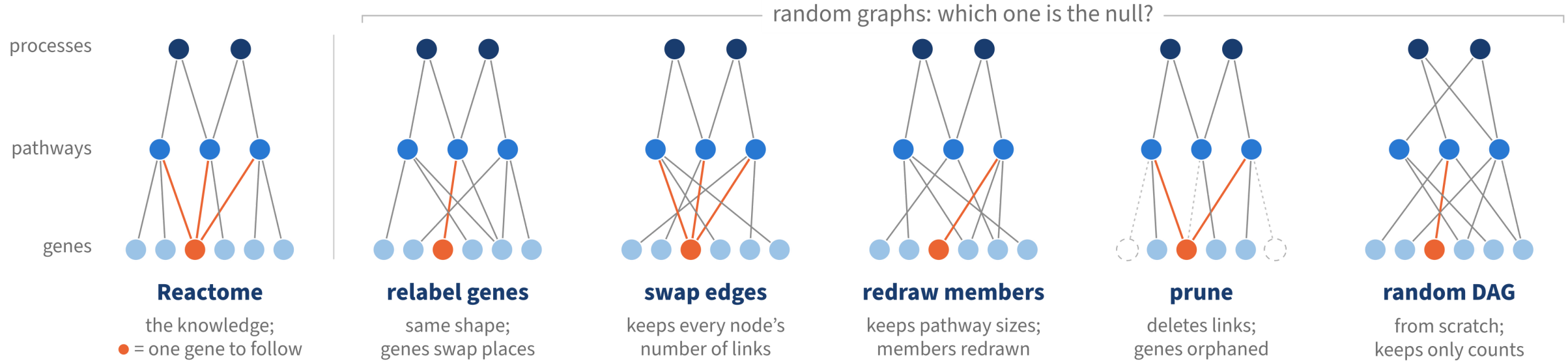
graph = input;
a vector per node,
one weight matrix per level

$$[(\mathbf{W} \odot \mathbf{M}) \mathbf{h}]_v = \sum_{u \rightarrow v} w_{uv} h_u \cdot \text{masked layer} = \text{one round of message passing}$$

hidden nodes have no data → **defined by position alone**

60% of P-NET's genes: same pathways as another gene

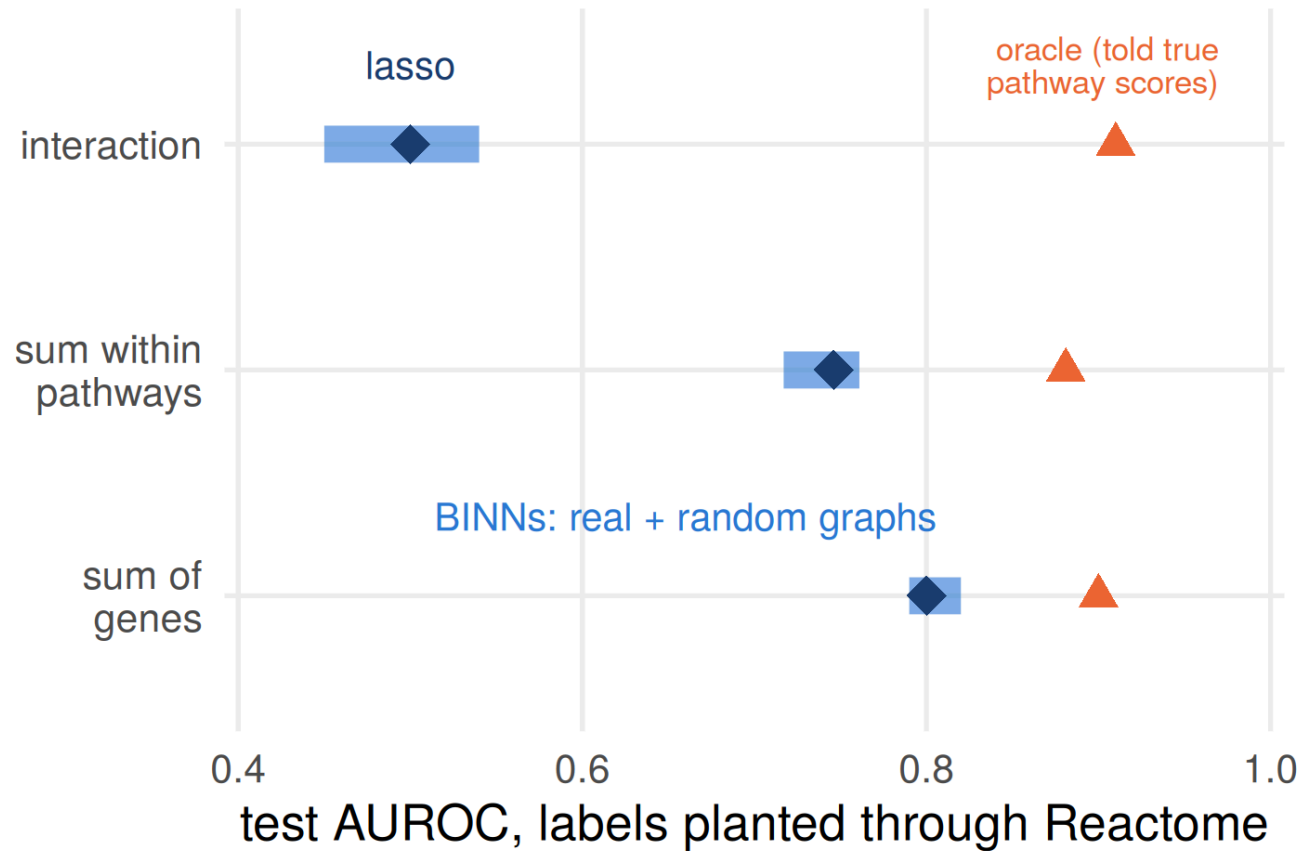
What is the null hypothesis?



H_0 : “*the knowledge is irrelevant*” · but **which** random graph?

straw-man null: reject it, claim the favoured alternative

Can't use it $\neq$ useless



- labels built **through** real pathways
- real $\approx$ random $\approx$ lasso $\ll$ oracle
- no power $\rightarrow$ **no conclusion**

ask instead

- equivalence, not “*n.s.*”
- nested models: knowledge as a **penalty**
- positive controls

McShane, Gelman et al. 2019, *Am. Stat.*

TCGA-PRAD RNA; Maani, Selby et al., in prep.

Every check is a new codebase

graph baked into code

- new null → new model
- new database → new model
- new data type → new paper

ecosystem gap

- BINNs: Python, one repo each
- R / Bioconductor: TCGA, `MultiAssayExperiment`, `limma`, `survival` ...
no BINNs

graph as **input** · checks as **one line**

binnr

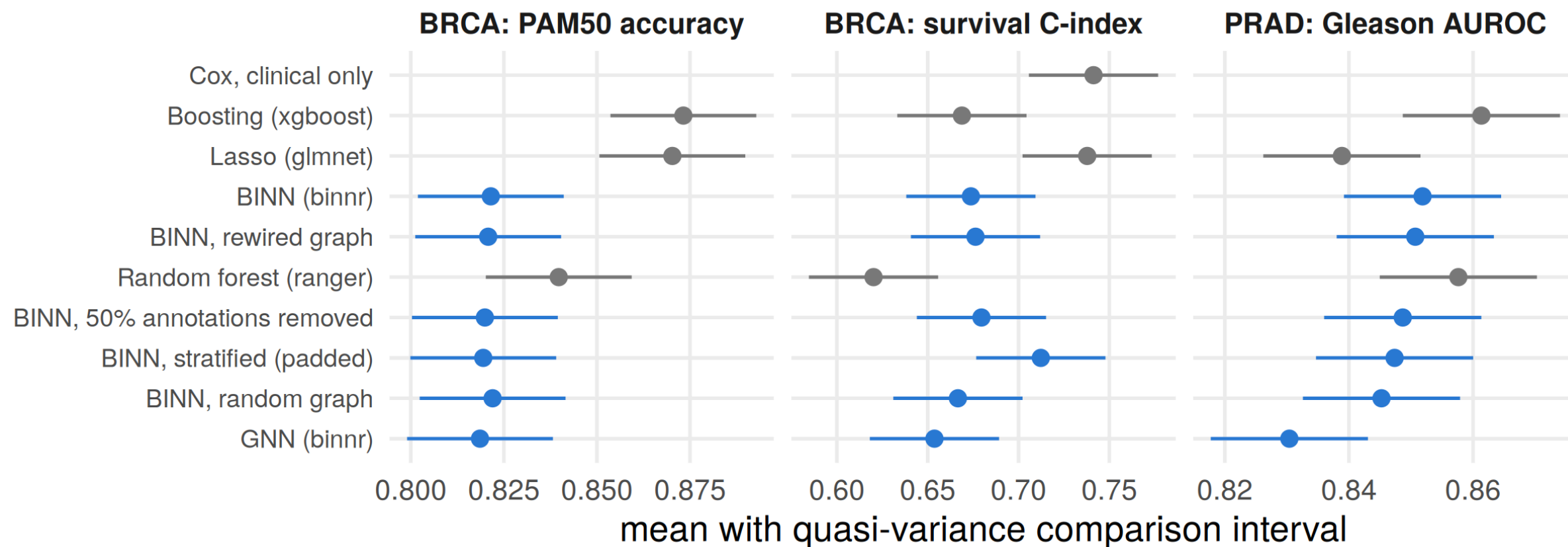
```
library(binnr)
brca <- tcga_cohort("BRCA")           # curatedTCGAData, cached
g    <- reactome_graph()              # igraph DAG; any gene sets work

fit  <- bn_fit(Surv(os_time, os_event) ~ age + stage, data = brca$clinical,
               omics = brca$omics, graph = g, arch = bn_arch("binn"))

update(fit, graph = bn_rewire(g, "degree")) # null graphs: one argument
update(fit, arch  = bn_arch("gnn", hidden_dim = 4))
node_activations(fit, new_omics)           # patients × nodes
```

torch · igraph · MultiAssayExperiment · formulas · tidy output · classification / regression / Cox

What it shows



real vs random graphs: **no detectable difference** · lasso matches or beats BINNs · age + stage hard to beat on survival

TCGA BRCA (n = 960) & PRAD (n = 491); 5×5-fold CV, tuned within folds. Also reproduces P-NET's split and Hartman et al.'s proteomics.

Next — and who wants in?

next

- knowledge as **prior**, not mask
- hierarchy nulls · random DAGs
- missing modalities · covariates
- cardiovascular cohorts
- CRAN · Bioconductor

let's talk: **data · nulls · use cases**



github.com/datasciapps/binnr

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